

Translated

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Modular Multiplicative Inverse

Definition

A [modular multiplicative inverse](#) of an integer a is an integer x such that $a \cdot x$ is congruent to 1 modulo some modulus m . To write it in a formal way: we want to find an integer x so that

$$a \cdot x \equiv 1 \pmod{m}.$$

We will also denote x simply with a^{-1} .

We should note that the modular inverse does not always exist. For example, let $m = 4$, $a = 2$. By checking all possible values modulo m , it should become clear that we cannot find a^{-1} satisfying the above equation. It can be proven that the modular inverse exists if and only if a and m are relatively prime (i.e. $\gcd(a, m) = 1$).

In this article, we present two methods for finding the modular inverse in case it exists, and one method for finding the modular inverse for all numbers in linear time.

Finding the Modular Inverse using Extended Euclidean algorithm

Consider the following equation (with unknown x and y):

$$a \cdot x + m \cdot y = 1$$

This is a [Linear Diophantine equation in two variables](#). As shown in the linked article, when $\gcd(a, m) = 1$, the equation has a solution which can be found using the [extended Euclidean algorithm](#). Note that $\gcd(a, m) = 1$ is also the condition for the modular inverse to exist.

Now, if we take modulo m of both sides, we can get rid of $m \cdot y$, and the equation becomes:

$$a \cdot x \equiv 1 \pmod{m}$$

Thus, the modular inverse of a is x .